Two Real-World Innovation Challenges

Mobile operations, automated vehicle detection, and AI-powered insights from parking bookings.

About GoPark

GoPark helps drivers find, reserve, and pay for parking through its mobile platform. These challenges invite AUB students to prototype solutions to operational and engagement problems with practical potential for deployment.

CHALLENGE 01 | Smart Parking Attendant & Automated Check-In/Check-Out

Mobile development • IoT • UX • Automation

The problem

Parking operators need reliable information about when booked vehicles arrive and leave. Manual updates can cause inaccurate occupancy, missed check-ins, and delayed space availability. The central challenge is to detect real arrivals and departures with minimal intervention.

What to build

A mobile prototype for parking attendants, combined with an innovative driver-assisted or automated entry/exit confirmation mechanism.

Core capability	Expected behavior
Parking overview	Display available, reserved, and occupied spaces.
Vehicle check-in / check-out	Validate booking and log actual entry and exit timestamps.
Plate / booking lookup	Find bookings by plate number, booking ID, or QR code.
Operational alerts	Flag upcoming arrivals, overstays, and unconfirmed departures.
Activity history	Show attendant actions and changes to vehicle status.
Resilience	Handle incorrect scans, missed events, and intermittent connectivity.

The innovation question

How can GoPark know a car arrived or departed without depending entirely on attendants?

- Explore QR verification, GPS/geofencing, plate recognition, Bluetooth/IoT, or a hybrid method.
- Design driver incentives, such as loyalty points, discounts, or useful in-app benefits, to encourage reliable confirmation.

- Address false detection, privacy, cost, ease of use, and deployment at different parking locations.

Expected outcome

A demonstrable attendant workflow plus a proposed or working detection mechanism, with a brief explanation of its accuracy, cost, limitations, and incentive design.

CHALLENGE 02 | AI-Powered Booking Pattern Recognition

Artificial intelligence • Machine learning • Predictive analytics

The problem

Drivers exhibit different parking routines: daily weekday bookings, specific recurring days, multiple locations, and occasional usage. GoPark needs a scalable way to identify those routines from booking records only and turn them into structured insights for future notification decisions.

What to build

An AI engine that takes a dataset of historical bookings as input and produces a per-user pattern table. It should identify recurring days and times, preferred locations, likely next booking, and prediction confidence.

Input: booking records only

Field	Meaning
User ID / Booking ID	Identifiers for the user and reservation.
Parking location	Location or parking ID booked.
Booking date / start time	When the reservation is scheduled to begin.
End time	When the reserved parking period ends.
Booking status	Completed, cancelled, or other applicable status.

Required output: user pattern table

User	Detected pattern	Time / place	Next likely booking	Confidence
U001	Mon / Wed / Fri	8 AM · AUB	Next Monday, 8 AM	92%
U002	Weekdays	9 AM · BDD	Next weekday, 9 AM	95%
U003	Tue / Thu	10 AM · LAU	Next Tuesday, 10 AM	87%
U004	Occasional	Variable · Hamra	Not predictable	35%

Illustrative values only. The engine must calculate its own patterns and confidence scores using test data.

Main questions for teams

- How can the engine distinguish repeatable habits from coincidence?
- Can it detect multiple patterns for a single user and changing routines over time?

- How should it handle sparse histories, cancellations, and inconsistent booking times?
- When does confidence become too low to recommend a next-booking reminder?

Recommended additional fields

Include last booking date, booking frequency, detected inactivity, recommended reminder timing, and an interpretable reason for each prediction. Reminder timing is an output recommendation, not a requirement to build a notification-sending system.

Expected outcome

A working model or rule-based/ML hybrid that ingests anonymized or synthetic bookings, exports its results in CSV/JSON or through an API, and demonstrates predictions against held-out booking history.

Hackathon Deliverables & Evaluation

The goal is a credible and testable prototype, not a production-ready platform. Teams may demonstrate their ideas using synthetic or anonymized datasets and simulated integrations.

Expected deliverables

- Challenge 1: Attendant mobile workflow, arrival/departure mechanism, edge-case handling, and proposed driver incentive.
- Challenge 2: Booking data ingestion, pattern table, prediction logic, machine-readable export, and basic evaluation.
- For both: Short demo, simple architecture explanation, and list of next steps for real deployment.

Suggested evaluation criteria

Criterion	Weight
Innovation and creativity	25%
Technical implementation	25%
Practical feasibility and scalability	20%
User experience and output clarity	15%
Potential value for GoPark	15%

Potential collaboration

GoPark can support teams with problem context, sample user journeys, synthetic or appropriately anonymized data where available, and feedback from its technical and operations teams. Promising prototypes may be considered for subsequent collaboration by mutual agreement.

Our objective: Turn real parking and mobility challenges into opportunities for AUB students to build solutions with measurable real-world potential.

GoPark | Book your parking, stress-free